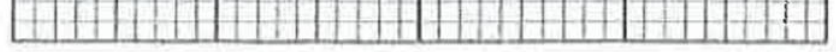
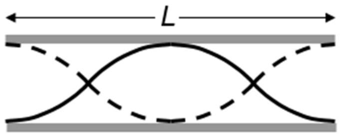
	
4(a)	The stationary wave is formed by the superposition of two progressive waves of equal amplitude, frequency and speed but travelling in opposite directions.
(b)	The representation shows the amplitude of oscillation of air molecules parallel to the axis of the pipe. The amplitude of oscillation is maximum (distance between the two curves is maximum) at the top of the tube and zero (the two curves meet and distance between them is zero) at the surface of the water.
(c)	
(d)	 For first harmonic shown in (c), $\lambda = 2L$ Speed of wave $v = f\lambda = 540 \times 2L = 1080L$ For second harmonic shown in (d), $\lambda' = L$ Since speed of wave is constant, $f'L = 1080L$ $f' = 1080$ Hz Hence the next highest frequency is 1080 Hz.
5(a)	